

`lcd_init()` configures and initialises the LCD.

`adc_init()` initialises the built-in ADC of AT-Mega16.

`rotate_motor()` rotates the stepper motor in clockwise or counter-clockwise direction as per the entered angle value till the motor reaches the required position.

Construction and testing

Actual-size, single-side PCB layout of the joystick based stepper motor angle rotation controller using AVR MCU is shown in Fig. 3 and its component layout in Fig. 4. Actual-size, single-side PCB layout of the power supply section is shown in Fig. 5 and its component layout in Fig. 6.

Use any 2-wire electrical cable between CON1 and CON2 to connect 5V power supply to the controller circuit. When the circuit is first switched on, the LCD displays the angle as 0°. The motor also rests at 0°. To set the desired angle, rotate the shaft of the joystick potmeter. As potmeter shaft is rotated counter-clockwise (increase) or clockwise (decrease), the angle is increased or decreased in steps of 15° and is displayed on the LCD. As soon as the angle is changed, the stepper motor

starts rotating in order to reach the new angle. For example, if the angle

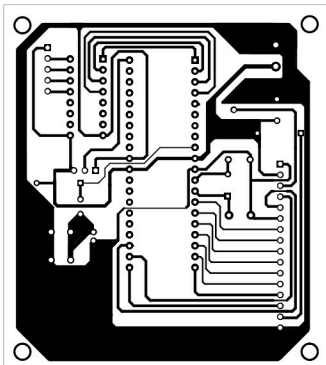


Fig. 3: Actual-size PCB layout of the joystick based stepper motor angle controller using AVR MCU

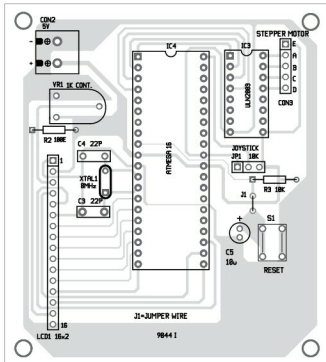


Fig. 4: Component layout of the PCB shown in Fig. 3

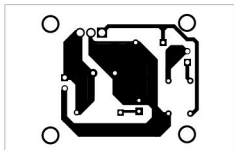


Fig. 5: Actual-size PCB layout of the power supply

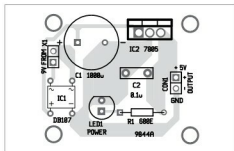


Fig. 6: Component layout of the power supply shown in Fig. 5

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at source.efymag.com

is set to 30° using the potmeter, the motor immediately starts rotating and stops at 30°. If the angle is increased by rotating the potmeter in counter-clockwise direction, the motor also rotates in the same direction. Similarly, if the angle is decreased by rotating the potmeter in clockwise direction, the motor rotates clockwise.

EFY note. The circuit has been checked in EFY Lab with a regular potentiometer instead of a joystick. ●



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